

over the Internet with at least one other student in your class:

$$\forall x(I(x) \Rightarrow \exists y((x \neq y) \wedge C(x, y)))$$

k) Someone in your class has an Internet connection, but has not chatted with anyone else in your class:

$$\exists x(I(x) \wedge \forall y((x \neq y) \Rightarrow \neg C(x, y)))$$

l) There are two students in your class who have not chatted with each other over the Internet:

$$\exists x \exists y (\neg C(x, y) \wedge (x \neq y) \wedge \forall z \forall w ((z \neq w) \wedge \neg C(z, w) \Rightarrow (z = x \wedge w = y) \vee (z = y \wedge w = x)))$$

m) There is a student in your class who has chatted with everyone in your class over the Internet: $\exists x \forall y C(x, y)$

n) There are at least two students in your class who have not chatted with the same person in your class: $\exists x \exists y \exists z ((x \neq y) \wedge \neg C(x, z) \wedge \neg C(y, z))$

(Imp) $\rightarrow$
o) There are two students in the class who between them have chatted with everyone else in the class: $\exists x \exists y \forall z ((x \neq z) \wedge (y \neq z) \Rightarrow (C(z, x) \vee C(z, y)))$

This statement means: There exist two particular students such that every other student in the class has chatted with ~~exactly one~~ at least one of these two students.

13) Let $M(x, y)$ be "x has sent y an email message" and $T(x, y)$ be "x has telephoned y," where the domain consists of all students in your class. Use quantifiers to express each of these statements. (Assume that all email messages that were sent are received, which is not the way things often work.)

a) Chou has never sent an email message to Koko: $\neg M(\text{Chou}, \text{Koko})$

b) Arlene has never sent an email message to or telephoned Sarah: $\neg M(\text{Arlene}, \text{Sarah}) \wedge \neg T(\text{Arlene}, \text{Sarah})$ (Imp)